

9th Class Physics – Past Paper 2024

Punjab Board (Rawalpindi Board)

Group-I (Paper Code: 5471)

Objective Type

Marks: 12

Time: 15 Minutes

Multiple Choice Questions (MCQs)

Q.No.	Question	A	B	C	D
1	Correct equation of motion	$V_f - V_i = at$ ✓	$V_i - V_f = at$	$V_f^2 - V_i^2 = a^2t^2$	$V_f^2 + V_i^2 = a^2t^2$
2	Significant figures in 0.0301	3 ✓	2	4	1
3	Unit of force	Ns	Kgms ⁻¹	Kgms⁻² ✓	Ns ⁻¹
4	$F_x = 6N$, $F_y = 8N$. Magnitude of F	48N	14N	10N ✓	100N
5	Gravitational field strength near Earth's surface	Zero	6.67 Nkg ⁻¹	10 Nkg⁻¹ ✓	8.9 Nkg ⁻¹
6	A 2 kg body on a 100 cm high table has KE =	50J	20J ✓	400J	Zero
7	Graph that does NOT obey Hooke's Law	(A)	(B)	(C) ✓	(D)
8	Water freezes at	32°F ✓	180°F	212°F	0°F
9	Metals are good conductors due to	Free electrons ✓	Big molecules	Small molecules	Rapid vibrations
10	$F = 20N$, $a = 2.5 \text{ ms}^{-2}$. Mass = ?	10kg	50kg	8kg ✓	0.25kg
11	Normal human body temperature	310K	273K	37K	98.6°F ✓

Q.No.	Question	A	B	C	D
12	Unit Nm (for torque) can also be used for	Power	Efficiency	Momentum	Work ✓

Subjective Type

Marks: 48

Time: 1 Hour 45 Minutes

Section – I

Question 2

Write short answers of any FIVE.

1. What is the International System of Units (SI)?
2. Define least count of vernier callipers.
3. Why do we need to measure extremely small intervals of time?
4. Write down two advantages of friction.
5. What will happen if all friction suddenly disappears?
6. What is meant by an isolated system?
7. Why does the value of "g" vary from place to place?
8. What is Global Positioning System (GPS)?

Question 3

Write short answers of any FIVE.

1. Draw a distance-time graph which shows variable speed.
2. Differentiate between speed and velocity.
3. How can vector quantities be represented graphically?
4. Define deforming force.
5. Can we use a hydrometer to measure the density of milk?
6. Define elasticity and give an example.
7. Differentiate between land breeze and sea breeze.
8. Why does transfer of heat in fluids take place by convection?

Question 4

Write short answers of any FIVE.

1. Define centre of gravity.
 2. Differentiate between like and unlike forces.
 3. Why is the height of vehicles kept as low as possible?
 4. Define power and write its SI unit.
 5. Differentiate between sound energy and light energy.
 6. Why are fossil fuels called non-renewable forms of energy?
 7. Define internal energy.
 8. Define evaporation.
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Section – II

Attempt any TWO questions.

Question 5

(a)

What are measuring instruments? Explain meter rod in detail.

(4 Marks)

(b)

How much time is required to change a momentum of **22 Ns** by a force of **20 N**?

(5 Marks)

Question 6

(a)

With the help of a speed-time graph, prove that:

$$2as = v_f^2 - v_i^2$$

(4 Marks)

(b)

A student presses her palm by her thumb with a force of **75 N**.

Contact Area = 1.5 cm^2

Find the pressure under the thumb.

(5 Marks)

Question 7

(a)

Define resolution of force. How can a force be resolved into its perpendicular components?

(4 Marks)

(b)

A block weighing **20 N** is lifted **6 m** vertically upward.

Calculate the potential energy stored in it.

(5 Marks)

End of Paper